🔴 RS002 Week 9 — English Worksheet

Topic: Dictionaries — Real Pokémon Data · **Course:** Pokédex App · **Time:** about 45 minutes

This week you think about the **dictionary** — a way to keep all of one Pokémon's facts together with **named keys** instead of four loose parallel lists. `pikachu["name"]` reads the value behind the key `"name"`. No more lining up indexes! In this worksheet you read code, debug on paper, and explain the dictionary code you wrote in your live lesson.

> 🧠 Words to know: dictionary, key, value, KeyError, list of dictionaries

1 · Recall 🧠

From memory:

Write four parallel lists holding the name, type, HP, and attack of two Pokémon, then print the first Pokémon's type.

2 · Predict 🔮

Read each block. Write what you think it would print.

```
pikachu = {"name": "피카츄", "type": "전기", "hp": 35}
print(pikachu["name"])
print(pikachu["hp"])
```

What two lines print?

```
pikachu = {"name": "피카츄", "type": "전기", "hp": 35}
print(pikachu["attack"])
```

There is no `"attack"` key. What happens with this block?

```
pokedex = [
    {"name": "피카츄", "type": "전기"},
    {"name": "이상해씨", "type": "풀"},
]
print(pokedex[1]["name"])
```

Two steps: first `[1]`, then `["name"]`. What gets printed?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it, then explain the fix.

Bug A — This should print the Pokémon's type. Right now it uses a list index on a dictionary and crashes.

```
pikachu = {"name": "피카츄", "type": "전기", "hp": 35}
print(pikachu[1])
```

Hint: dictionaries are read by **key name**, not by position number.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should print HP, but the key name has a typo, so it crashes with a `KeyError`.

```
pikachu = {"name": "피카츄", "type": "전기", "hp": 35}
print(pikachu["HP"])
```

Hint: keys are case-sensitive — match them exactly.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should print the type of **every** Pokémon in the dex. Right now it prints the same one three times.

```
pokedex = [
    {"name": "피카츄", "type": "전기"},
    {"name": "이상해씨", "type": "풀"},
    {"name": "파이리", "type": "불"},
]

for pokemon in pokedex:
    print(pokedex[0]["type"])
```

Hint: use the loop variable `pokemon`, not `pokedex[0]`.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code 📖

Read this working dictionary search carefully.

```
pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35, "attack": 55},
    {"name": "이상해씨", "type": "풀", "hp": 45, "attack": 49},
    {"name": "파이리", "type": "불", "hp": 39, "attack": 52},
]

search = input("어떤 포켓몬을 찾으시나요? ").strip()

found = False
for pokemon in pokedex:
    if pokemon["name"] == search:
        print(f"\n=== {pokemon['name']} ===")
        print(f"타입: {pokemon['type']}")
        print(f"체력: {pokemon['hp']}")
        print(f"공격력: {pokemon['attack']}")
        found = True

if found == False:
    print("도감에서 찾을 수 없습니다.")
```

What is `pokedex` — and what is each item inside it?

The loop variable is `pokemon`. What does `pokemon["name"]` read on each pass?

What is the job of the `found` variable, and when does it become `True`?

If the user types a name that is not in the dex, which line runs and why?

Why is this dictionary version clearer than four parallel lists with `names[i]`, `types[i]`, and so on?

5 · Explain Your Lesson Code 🎥

In your live lesson you wrote dictionary code. Now explain **that** code in a short phone video. You may show it running. Teach it like the viewer is new, and try to use: **dictionary**, **key**, **value**, **KeyError**, **list of dictionaries**.

> 1. Show one Pokémon dictionary from your lesson and read its keys out loud.
> 2. Run your search and show the matching Pokémon's full info.
> 3. Explain why `pokemon["name"]` is clearer than `names[i]`.
> 4. Mention one `KeyError` you hit (or could hit) and how to avoid it.

Write what you will say in your video. Plan it here before you record.

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Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.